Yuanning Zhang

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Research Interests

Arithmetic geometry; de Rham–Witt cohomology; prismatic F-gauges; Brauer groups.

Education

- **Ph.D. in Mathematics**, Northwestern University, 2021–present (Expected Spring 2027). Advisor: Benjamin Antieau.
- **B.A. in Mathematics and Computer Science**, University of California, Berkeley, 2017–2021.

Publications and Preprints

- Y. Zhang, **The stacky approach to de Rham–Witt cohomology**. Preprint, 2026. arXiv:2610.28149.
- Y. Zhang, **Harder–Narasimhan filtration of torsion F-gauges**. Preprint, 2026. arXiv:2609.27089.
- Y. Zhang, **Higher dimensional dominoes in de Rham–Witt cohomology**. Preprint, 2026. arXiv:2607.26323.
- The 2020 Polymath Jr. REU "q-binomials and the Grassmannian group", **Filtering cohomology of ordinary and Lagrangian Grassmannians**. *Involve* **15** (2022), 271–288.

Research Visits

- **ICERM, Brown University**, Research cluster participant, *K3 Surfaces over Finite Fields*, October 19–23, 2026.
- **SLMath**, Stephen Della Pietra Program Associate Fellow, *Motivic Homotopy Theory: Connections and Applications*, August 17–December 18, 2026.
- **University of California, Berkeley**, NSF FRG meeting: *Higher Categorical Structures in Algebraic Geometry*, March 19–21, 2025.

Selected Talks

- October 29, 2026. *The stacky approach to de Rham–Witt cohomology*, Simons Collaboration Work-in-Progress Seminar.
- October 9, 2026. *Harder–Narasimhan filtration of torsion F-gauges*, Student Arithmetic Geometry Seminar, University of California, Berkeley.
- September 23, 2026. *Overview* (opening talk), Mini Workshop on q-de Rham Cohomology and Habiro Rings, SLMath.
- April 25, 2026. *From F-crystals to Hodge–Witt cohomology*, IU–Purdue Joint Workshop on Prismatic F-gauges, Purdue University, Indianapolis Campus.
- March 30, 2026. *Nygaard filtered prismatization in mixed characteristics*, IU–Purdue Arithmetic Geometry Learning Seminar.
- Winter 2025. *Brauer group of schemes*, Graduate Number Theory Seminar, Northwestern University.
- Fall 2024. *Knots and primes*, Graduate Student Seminar, Northwestern University.

- Fall 2023. Condensed Mathematics Seminar, University of Chicago.
- 2021. Joint Mathematics Meetings, presentation on the Polymath REU Grassmannian cohomology project.

Seminar and Workshop Organization

- Co-organizer, *Graduate Student Seminar*, SLMath, Fall 2026 (with Erika Beserra).
- Organizer, *Mini Workshops*, SLMath, Fall 2026, including *q-de Rham Cohomology and Habiro Rings*, September 23–25, 2026.
- Organizer, *Prismatic F-gauges Learning Seminar*, Northwestern University, Winter–Spring 2024.

Teaching

- Teaching Assistant, *Northwestern University*, 2022–2025. Assisted with thirteen undergraduate course offerings, including calculus, linear algebra, abstract algebra, graph theory, and number theory.
- Mentor, *Directed Reading Program*, Northwestern University, Spring 2025. Guided an undergraduate in commutative algebra and algebraic geometry through weekly meetings.

Professional Service

Referee: Compositio Mathematica.

Conferences and Workshops (Selected)

- Bootcamp for the 2025 Summer Research Institute in Algebraic Geometry, Colorado State University, July 8–12, 2025. Working-group mentor: Alexander Petrov.
- Simons Collaboration on Perfection in Algebra, Geometry, and Topology, Annual Meetings, New York City, 2024, 2025, and 2026.
- Arizona Winter School on representation theory of p-adic groups, 2025.
- Life after the Telescope Conjecture, SLMath, 2024.
- p-adic Geometry Special Year Conferences, IAS, 2023–2024.
- Arizona Winter School on Abelian Varieties, 2024.
- MSRI Summer School on Derived Algebraic Geometry, UC Berkeley, 2023.
- Recent Advances in Algebraic K-theory, IHES, 2023.
- Topological Hochschild Homology and Zeta Values, Masterclass, University of Copenhagen, 2023.

Skills and Languages

Programming: Python, Java, C.
Languages: Chinese (native); English (fluent); French (intermediate); Japanese (basic).